

Homework 1: Lambda Calculus — Solutions

Problem 1

No. Let $M = (\lambda x. y) z$ and $N = y$. Then $M \rightarrow N$ but $fv(M) = \{y, z\} \neq \{y\} = fv(N)$.

Problem 2

For convenience, we use the following shorthand notations:

$$\begin{aligned} M &= \lambda f. \lambda x. f (f x), \\ P &= \lambda b. \lambda x. \lambda y. b y x, \\ Q &= \lambda z. \lambda w. z. \end{aligned}$$

Then,

$$\begin{aligned} M P Q &\rightarrow (\lambda x. P (P x)) Q \\ &\rightarrow P (P Q) \\ &\rightarrow \lambda x. \lambda y. (P Q) y x \\ &\rightarrow \lambda x. \lambda y. (\lambda x. \lambda y. Q y x) y x \\ &\rightarrow \lambda x. \lambda y. (\lambda u. Q u y) x \\ &\rightarrow \lambda x. \lambda y. Q x y \\ &\rightarrow \lambda x. \lambda y. (\lambda w. x) y \\ &\rightarrow \lambda x. \lambda y. x \end{aligned}$$

Problem 3

For convenience, we use the following shorthand notation:

$$M = \lambda f. \lambda x. f (f x).$$

Then,

$$\begin{aligned} (\lambda f. f f) M &\rightarrow M M \\ &\rightarrow \lambda x. M (M x) \\ &\rightarrow \lambda x. M (\lambda z. x (x z)) \\ &\rightarrow \lambda x. \lambda y. (\lambda z. x (x z)) ((\lambda z. x (x z)) y) \\ &\rightarrow \lambda x. \lambda y. (\lambda z. x (x z)) (x (x y)) \\ &\rightarrow \lambda x. \lambda y. x (x (x (x y))) \end{aligned}$$

Problem 4

For convenience, we use the following shorthand notations:

$$\begin{aligned} I &= \lambda t. t, & K &= \lambda s. \lambda t. s, \\ J &= \lambda s. I s, & C &= \lambda s. s s s, \\ M &= \lambda t. C (t K). \end{aligned}$$

$$\begin{aligned}
(\lambda t. t J) M I &\rightarrow M J I \\
&\rightarrow C (J K) I \\
&\rightarrow (J K) (J K) (J K) I \\
&\rightarrow (I K) (J K) (J K) I \\
&\rightarrow K (J K) (J K) I \\
&\rightarrow (\lambda t. J K) (J K) I \\
&\rightarrow (J K) I \\
&\rightarrow (I K) I \\
&\rightarrow K I \\
&\rightarrow \lambda t. I \\
&= \lambda t. (\lambda t. t)
\end{aligned}$$

Problem 5

(a) Substitution

$$\begin{aligned}
(\text{let } x = M_1 \text{ in } M_2)[N/x] &\stackrel{\text{def}}{=} \text{let } x = M_1[N/x] \text{ in } M_2 \\
(\text{let } y = M_1 \text{ in } M_2)[N/x] &\stackrel{\text{def}}{=} \text{let } y = M_1[N/x] \text{ in } M_2[N/x], \text{ where } y \notin \text{fv}(N) \\
(\text{let } y = M_1 \text{ in } M_2)[N/x] &\stackrel{\text{def}}{=} \text{let } z = M_1[N/x] \text{ in } M_2[z/y][N/x], \text{ where } y \in \text{fv}(N) \text{ and } z \text{ fresh}
\end{aligned}$$

(b) Reduction

For convenience, we use the following shorthand notations:

$$\begin{aligned}
C &= \lambda n. \lambda m. \lambda f. \lambda x. n f (m f x), \\
B &= \lambda f. \lambda x. f (f x), \\
A &= \lambda f. \lambda x. f x.
\end{aligned}$$

Then,

$$\begin{aligned}
&\text{let } c = C \text{ in } (\text{let } b = B \text{ in } (\text{let } a = A \text{ in } c b a)) \\
&\rightarrow \text{let } b = B \text{ in } (\text{let } a = A \text{ in } C b a) \\
&\rightarrow \text{let } a = A \text{ in } C B a \\
&\rightarrow C B A \\
&\rightarrow (\lambda m. \lambda f. \lambda x. B f (m f x)) A \\
&\rightarrow \lambda f. \lambda x. B f (A f x) \\
&\rightarrow \lambda f. \lambda x. (\lambda x. f (f x)) (A f x) \\
&\rightarrow \lambda f. \lambda x. f (f (A f x)) \\
&\rightarrow \lambda f. \lambda x. f (f ((\lambda x. f x) x)) \\
&\rightarrow \lambda f. \lambda x. f (f (f x))
\end{aligned}$$

Common mistake. Reducing the innermost let first is invalid:

$$\begin{aligned}
&\text{let } c = C \text{ in } (\text{let } b = B \text{ in } (\text{let } a = A \text{ in } c b a)) \\
&\not\rightarrow \text{let } c = C \text{ in } (\text{let } b = B \text{ in } c b A)
\end{aligned}$$

There is no rule allowing reduction inside a let body; the following rule is *not* provided:

$$\frac{M_2 \rightarrow M'_2}{\text{let } x = M_1 \text{ in } M_2 \rightarrow \text{let } x = M_1 \text{ in } M'_2}$$